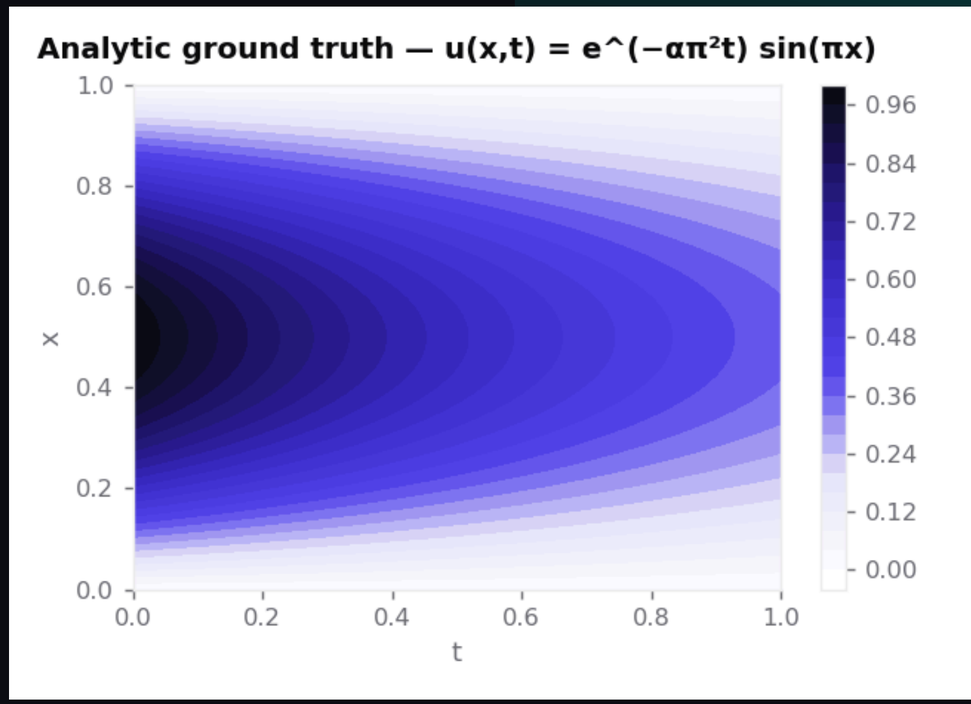
# ETH Hackathon 2026

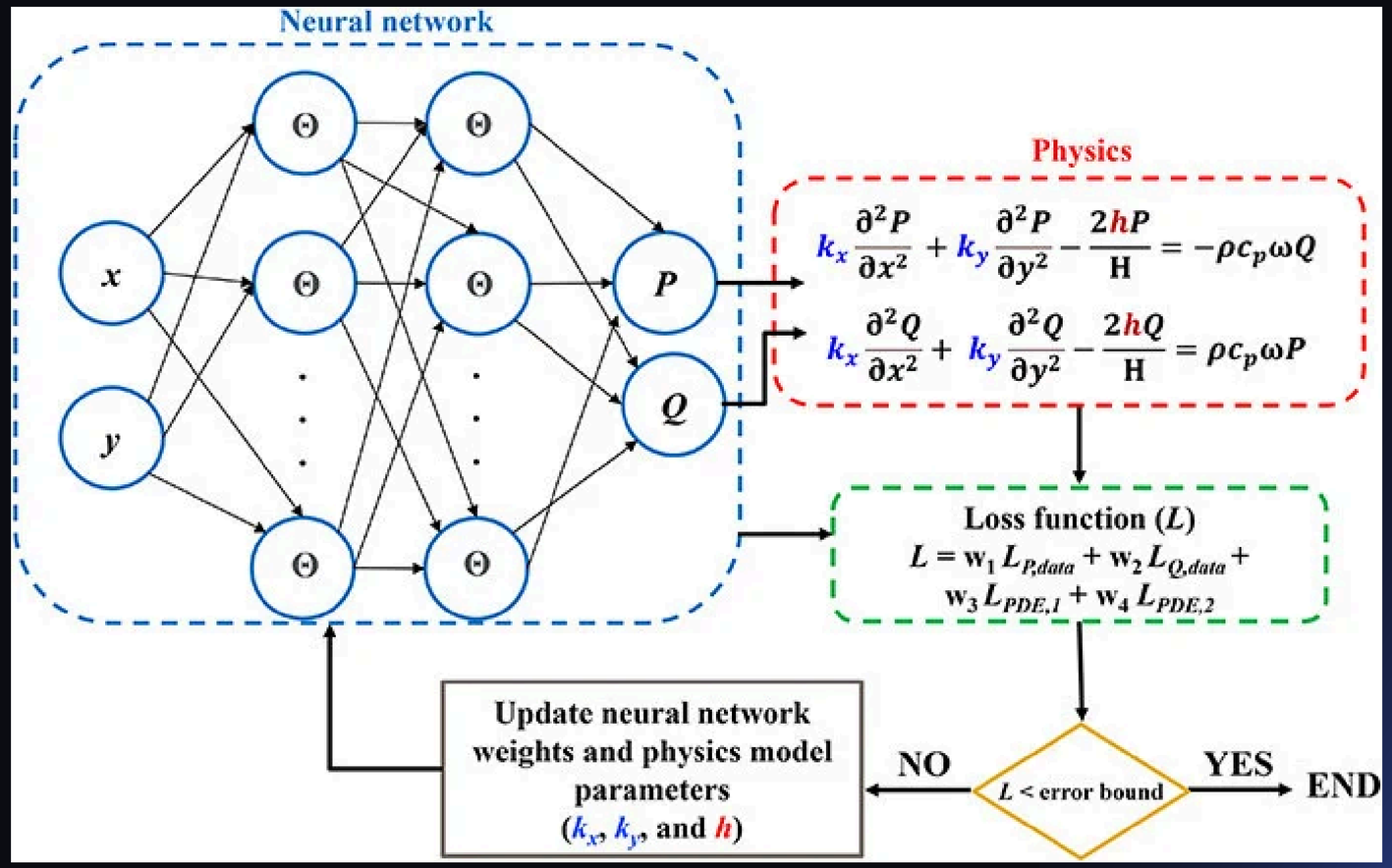
Quandela Challenge by PhotoFridge

# Objectives

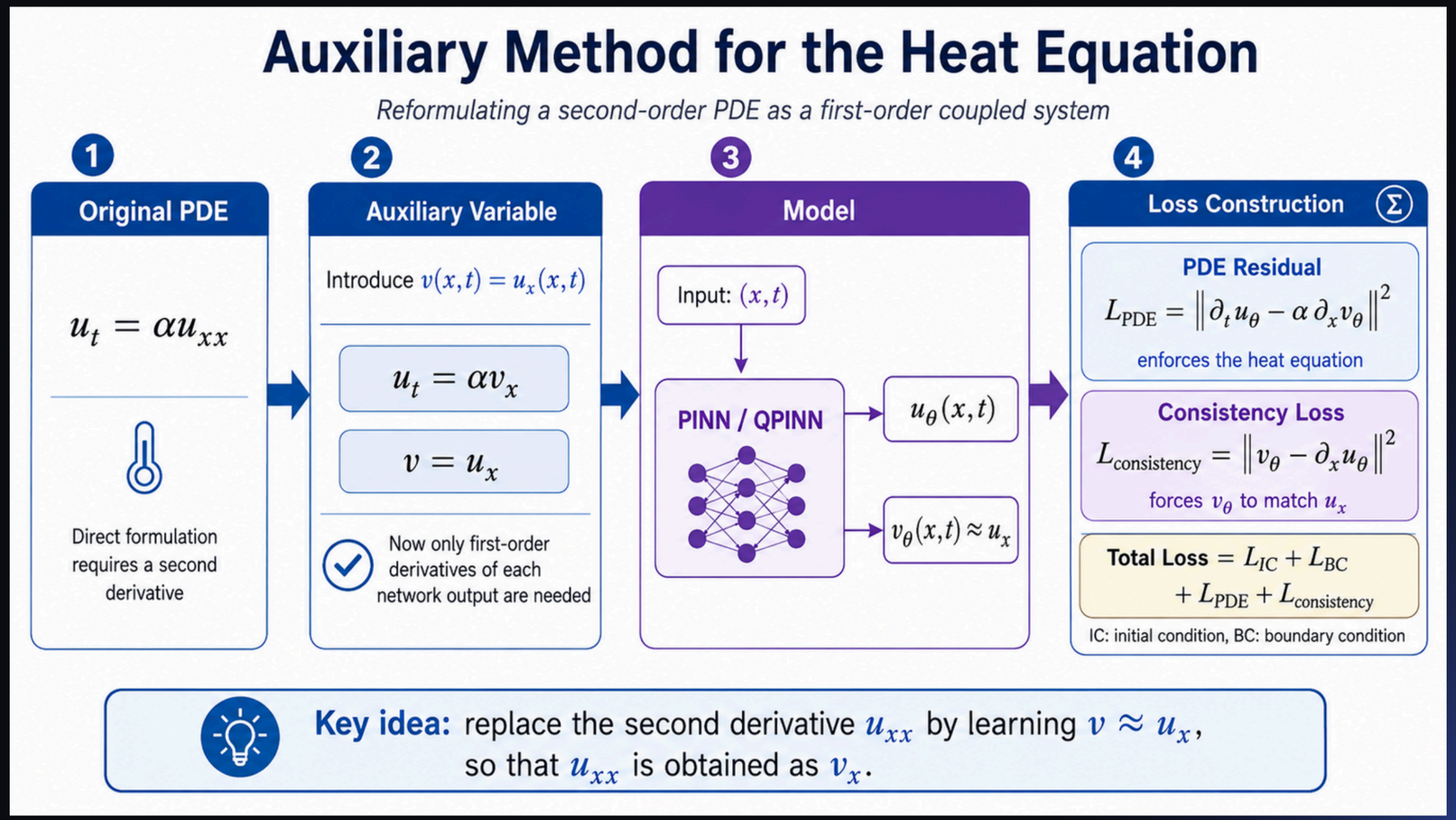
- Reproduce 1D heat equation
- Get a grasp of the models and try out different (Q-) NN architectures
- Compare and analyze the data
- Probe the quantum layer
- Push the models to their limits and look for quantum advantage



# Models

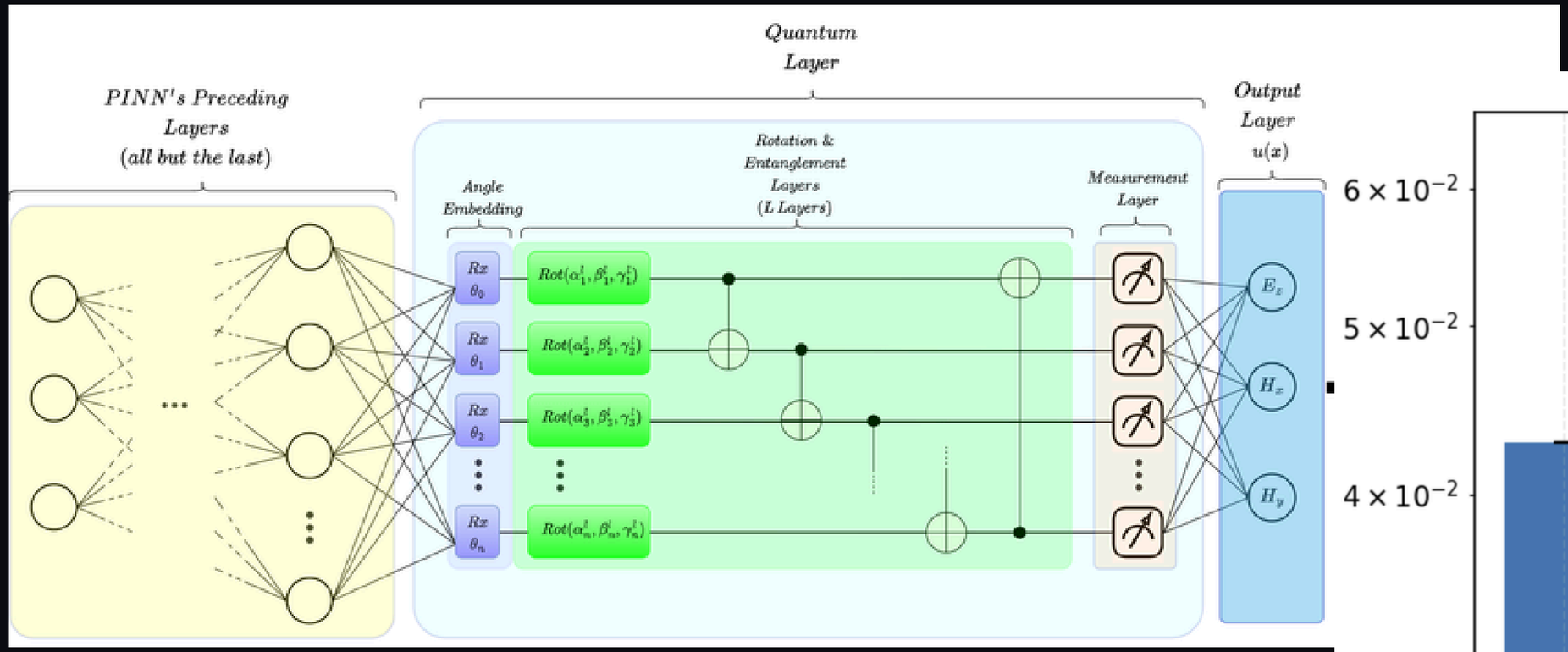

Neural network
x
y
Θ
P
Q
Physics
$k_x \frac{\partial^2 P}{\partial x^2} + k_y \frac{\partial^2 P}{\partial y^2} - \frac{2hP}{H} = -\rho c_p \omega Q$
$k_x \frac{\partial^2 Q}{\partial x^2} + k_y \frac{\partial^2 Q}{\partial y^2} - \frac{2hQ}{H} = \rho c_p \omega P$
Loss function ($L$)
$L = w_1 L_{P,data} + w_2 L_{Q,data} + w_3 L_{PDE,1} + w_4 L_{PDE,2}$
Update neural network weights and physics model parameters ($k_x$, $k_y$, and $h$)
NO
$L$ < error bound
YES
END

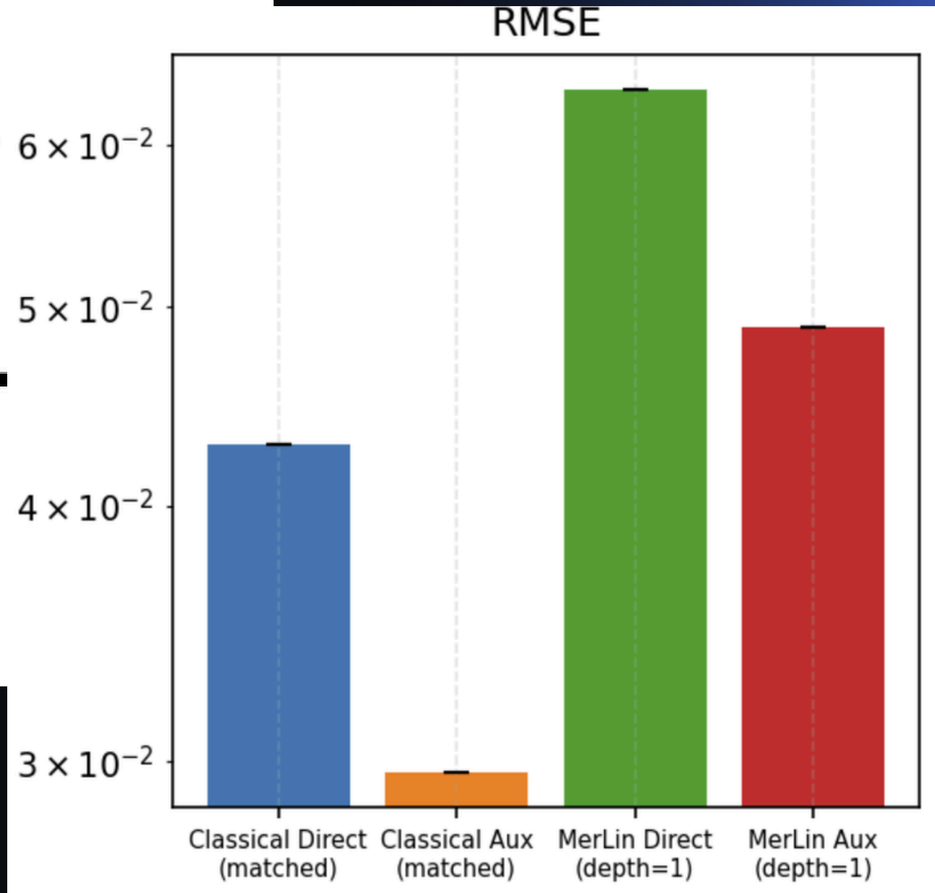
# Models


Auxiliary Method for the Heat Equation
Reformulating a second-order PDE as a first-order coupled system
1
Original PDE
$u_t = \alpha u_{xx}$
Direct formulation requires a second derivative
2
Auxiliary Variable
Introduce $v(x,t) = u_x(x,t)$
$u_t = \alpha v_x$
$v = u_x$
Now only first-order derivatives of each network output are needed
3
Model
Input: $(x,t)$
PINN / QPINN
$u_\theta(x,t)$
$v_\theta(x,t) \approx u_x$
4
Loss Construction
PDE Residual
$L_{\mathrm{PDE}} = \left\| \partial_t u_\theta - \alpha\, \partial_x v_\theta \right\|^2$
enforces the heat equation
Consistency Loss
$L_{\mathrm{consistency}} = \left\| v_\theta - \partial_x u_\theta \right\|^2$
forces $v_\theta$ to match $u_x$
Total Loss $= L_{IC} + L_{BC} + L_{\mathrm{PDE}} + L_{\mathrm{consistency}}$
IC: initial condition, BC: boundary condition
Key idea: replace the second derivative $u_{xx}$ by learning $v \approx u_x$, so that $u_{xx}$ is obtained as $v_x$.

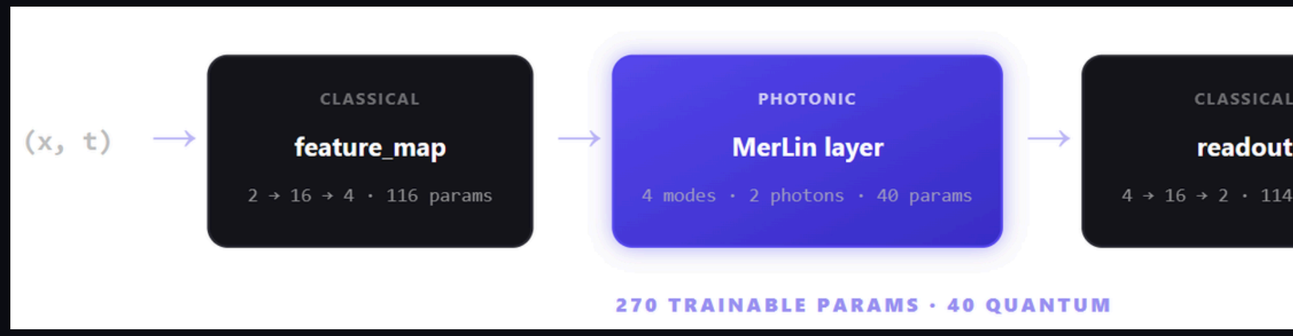
# Models

Introduce **trace loss**: to make sure quantum state is normalized



1) C_PINN (Classical PINN)




2) AC_PINN (Auxilarated Classical PINN)

3) AC_Q_PINN (Auxilarated Quantum PINN)

4) C_MLP_M (classical MLP mapped PINN)

MerLin layer replaced by tiny MLP of matched param count

5) Cust_Q_PINN (customised Quantum PINN)

6) H_Q_PINN (Hybrid Quantum PINN)

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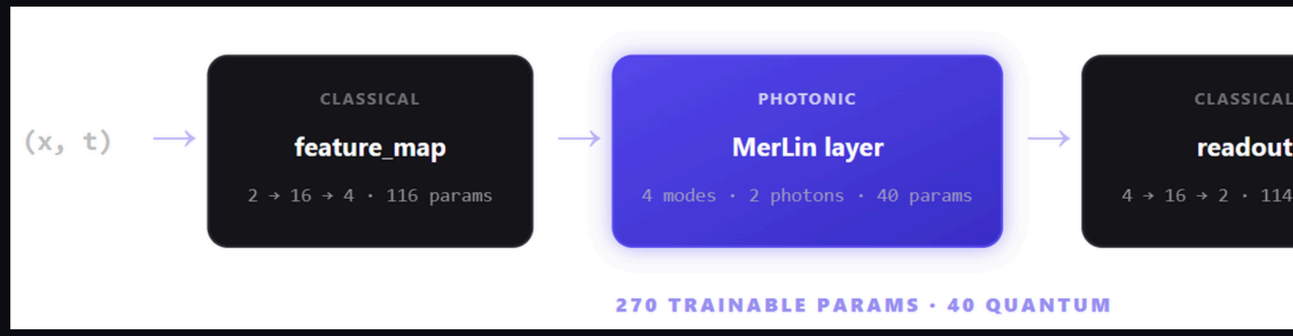


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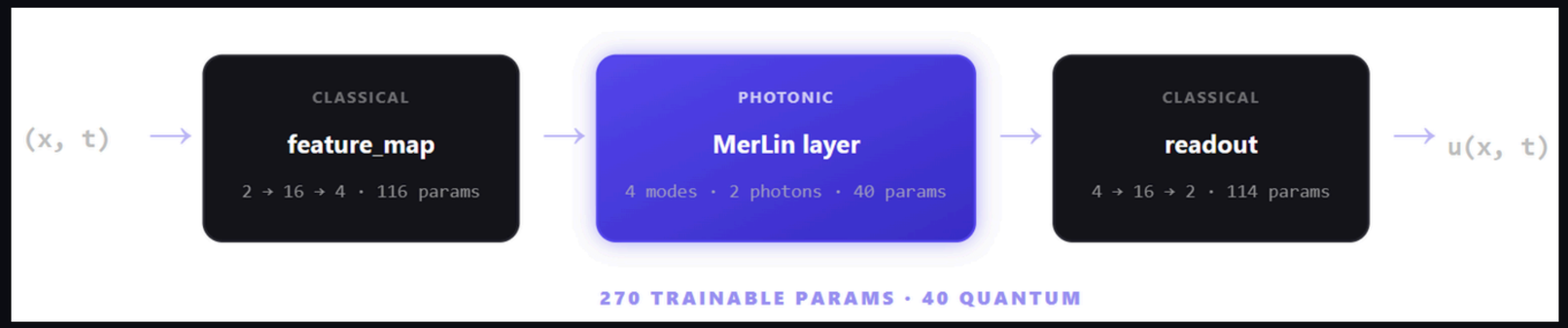


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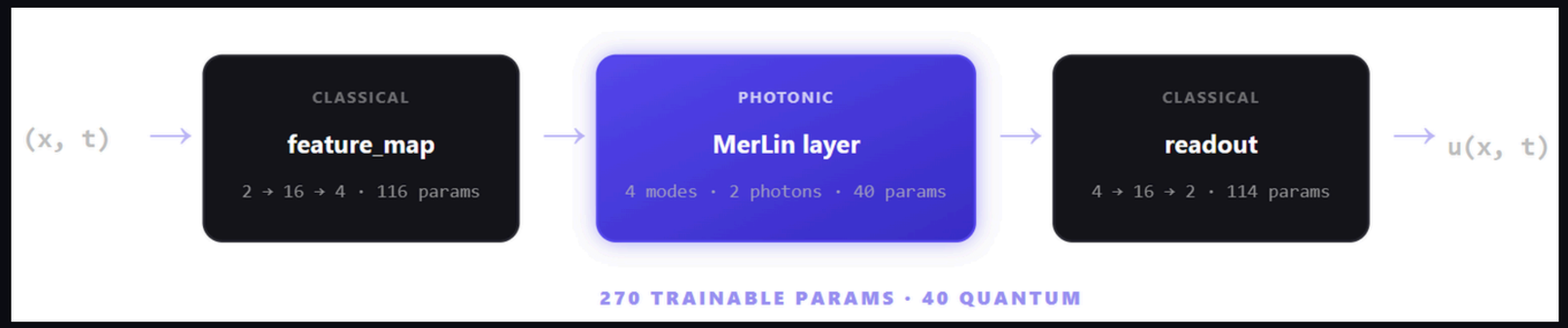


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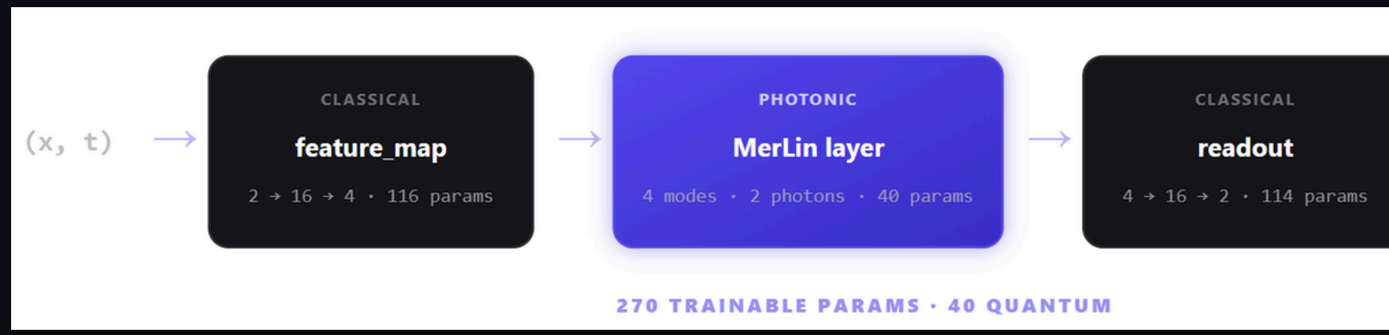
## 6) H_Q_PINN (Hybrid Quantum PINN)

1) C_PINN (Classical PINN)




2) AC_PINN (Auxilarated Classical PINN)

3) AC_Q_PINN (Auxilarated Quantum PINN)

4) C_MLP_M (classical MLP mapped PINN)

MerLin layer replaced by tiny MLP of matched param count

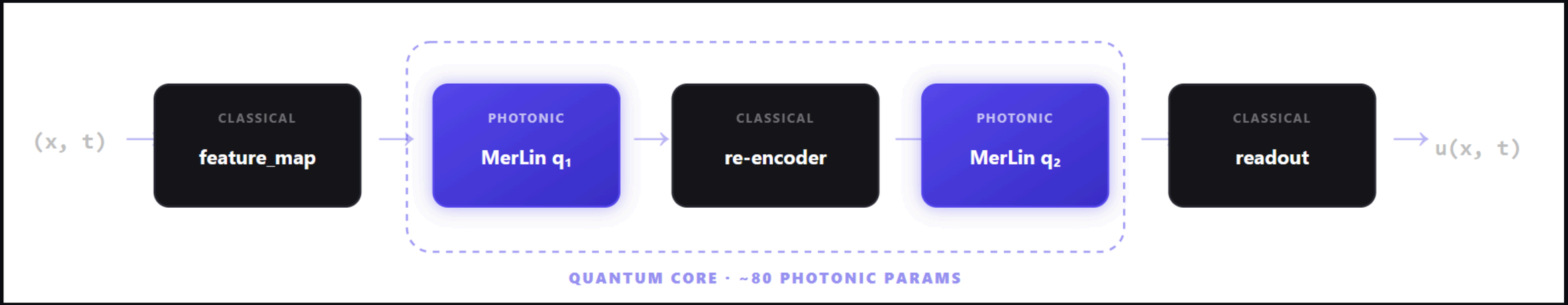
5) Cust_Q_PINN (customised Quantum PINN)

6) H_Q_PINN (Hybrid Quantum PINN)

# Classical flat MLP

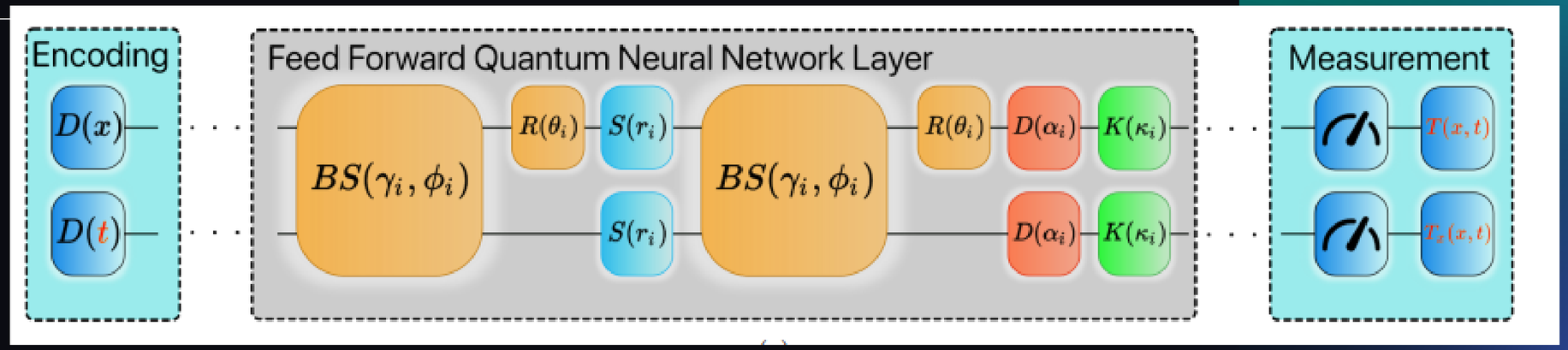
MerLin layer replaced by tiny MLP of matched param count

# enhanced QPINN with two stacked photonic layers and IC warm-up



# Custom Circuit

Cust_Q_PINN

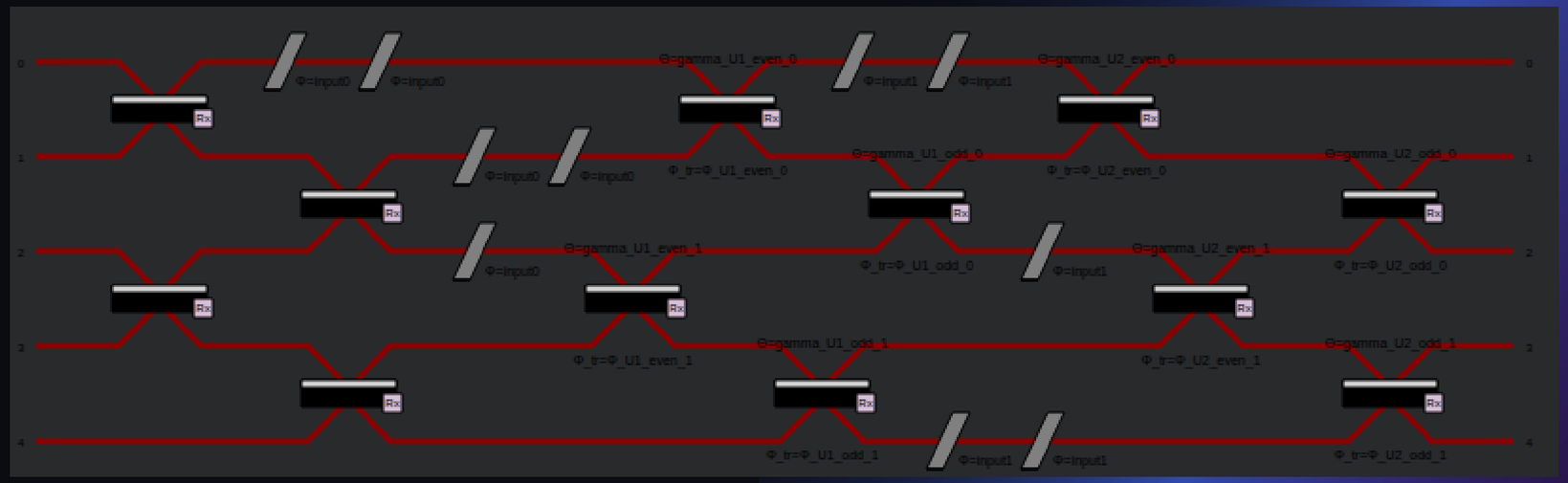


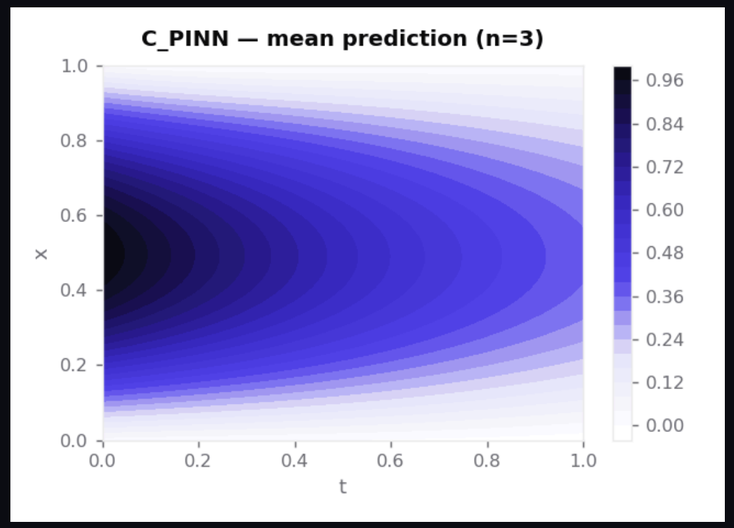

$$U_{\text{prep}}^{\text{fixed}}(\pi/2) \rightarrow PS_x(z_1(x,t)) \rightarrow U_1^{\text{trainable}}(\gamma,\phi) \rightarrow PS_t(z_2(x,t)) \rightarrow U_2^{\text{trainable}}(\gamma,\phi) \rightarrow p_{\text{count}}.$$

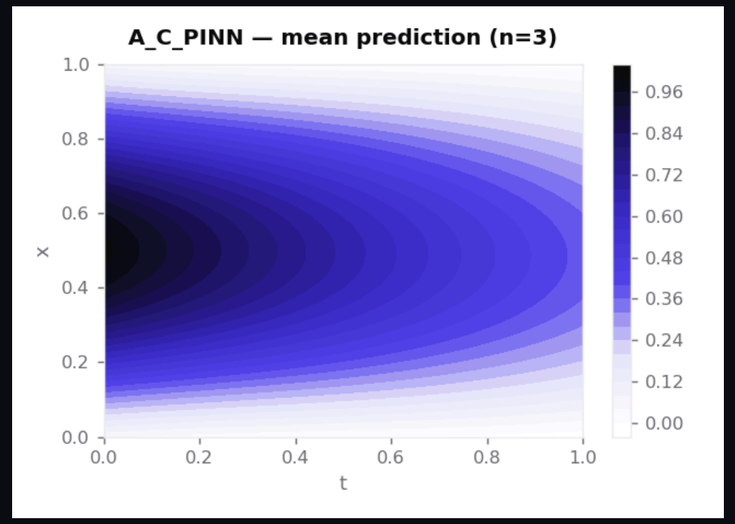
Orthogonal encoding (1,-1,0,-1,1)

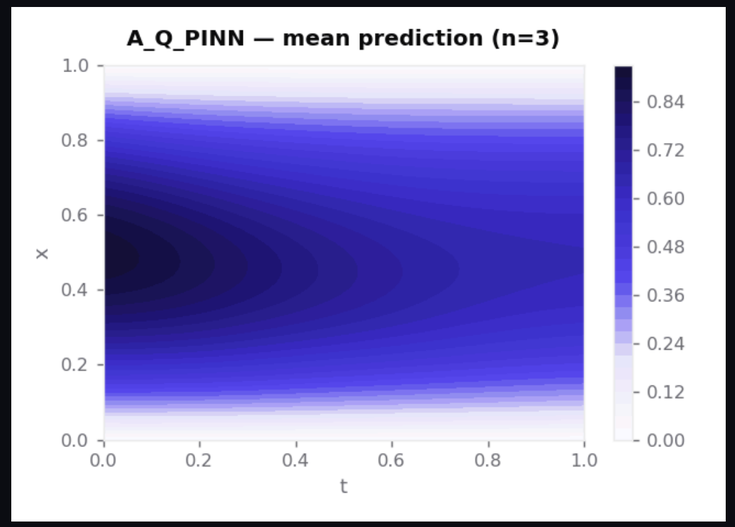
For five modes, three qubits:

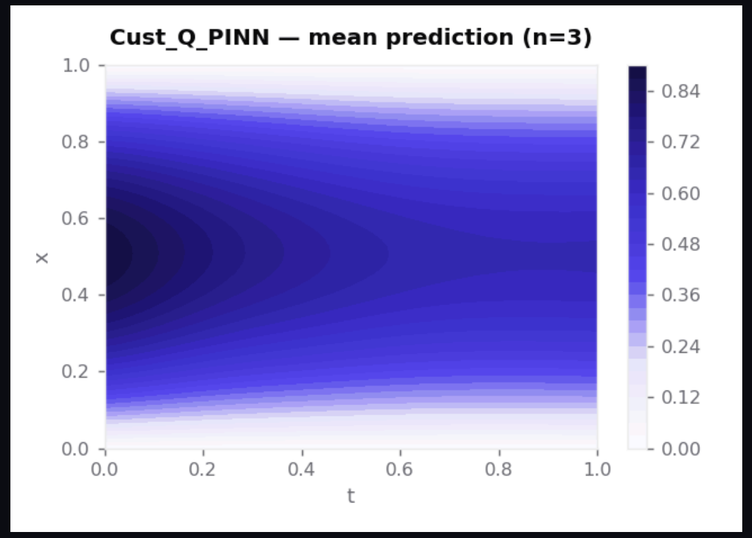
$$|1,1,1,0,0\rangle$$

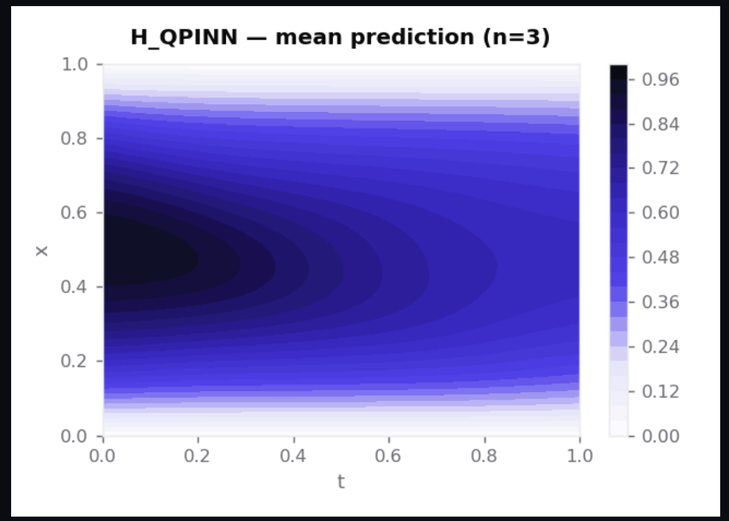


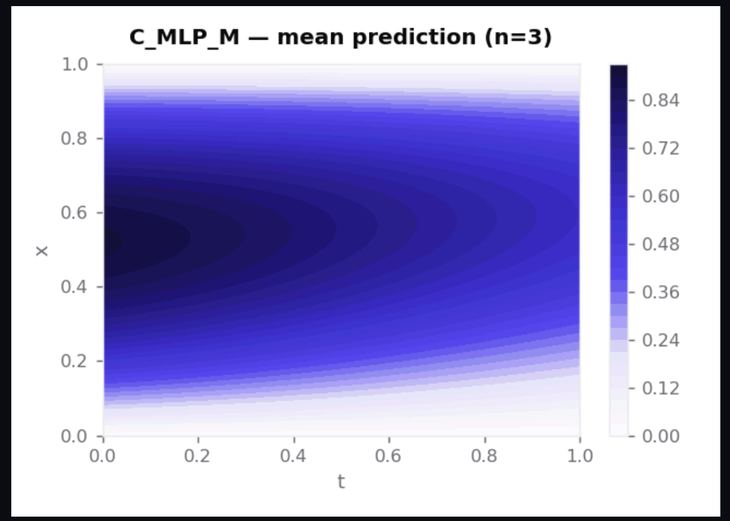

C_PINN — mean prediction (n=3)
x
t
0.0 0.2 0.4 0.6 0.8 1.0
0.00 0.12 0.24 0.36 0.48 0.60 0.72 0.84 0.96


A_C_PINN — mean prediction (n=3)
x
t
0.0 0.2 0.4 0.6 0.8 1.0
0.00 0.12 0.24 0.36 0.48 0.60 0.72 0.84 0.96


A_Q_PINN — mean prediction (n=3)
x
t
0.0 0.2 0.4 0.6 0.8 1.0
0.00 0.12 0.24 0.36 0.48 0.60 0.72 0.84


Cust_Q_PINN — mean prediction (n=3)
x
t
0.0 0.2 0.4 0.6 0.8 1.0
0.00 0.12 0.24 0.36 0.48 0.60 0.72 0.84


H_QPINN — mean prediction (n=3)
x
t
0.0 0.2 0.4 0.6 0.8 1.0
0.00 0.12 0.24 0.36 0.48 0.60 0.72 0.84 0.96


C_MLP_M — mean prediction (n=3)
x
t
0.0 0.2 0.4 0.6 0.8 1.0
0.00 0.12 0.24 0.36 0.48 0.60 0.72 0.84

| Model | Rel. $L^2$ (mean ± std) | Max abs error | PDE residual (mean) | Params | Time / seed |
|---|---|---|---|---|---|
| 2_C_PINN | 6.15e-3 ± 3.31e-3 | 6.85e-3 | 1.10e-1 | 3,297 | 16.6 s |
| 3_A_C_PINN | 7.44e-3 ± 2.17e-3 | 8.1e-3 | ~1.0e-1 | 3,298 | ~17 s |
| 4_A_Q_PINN | 4.83e-3 ± 1.79e-3 | 5.0e-3 | 1.06e-1 | 270 (40 quantum) | 189.6 s |
| 5_C_MLP_M | 4.41e-3 ± 1.54e-3 | 4.7e-3 | 9.7e-2 | 270 (0 quantum) | 12.6 s |
| 6_Cust_Q_PINN | 3.31e-3 ± 2.15e-3 | 3.7e-3 | 1.01e-2 | varies w/ n_modes | — |
| 7_H_QPINN | 1.32e-3 ± 6.31e-4 | 1.4e-3 | 1.5e-2 | 4,906 (80 quantum) | 390.8 s |

- Relative errors very comparable across the board
- Quantum shows advantage in parameter
- seed can play a big role
- Runtime has to be looked at in further detail
- different models give different insights into quantum layers

# Methods

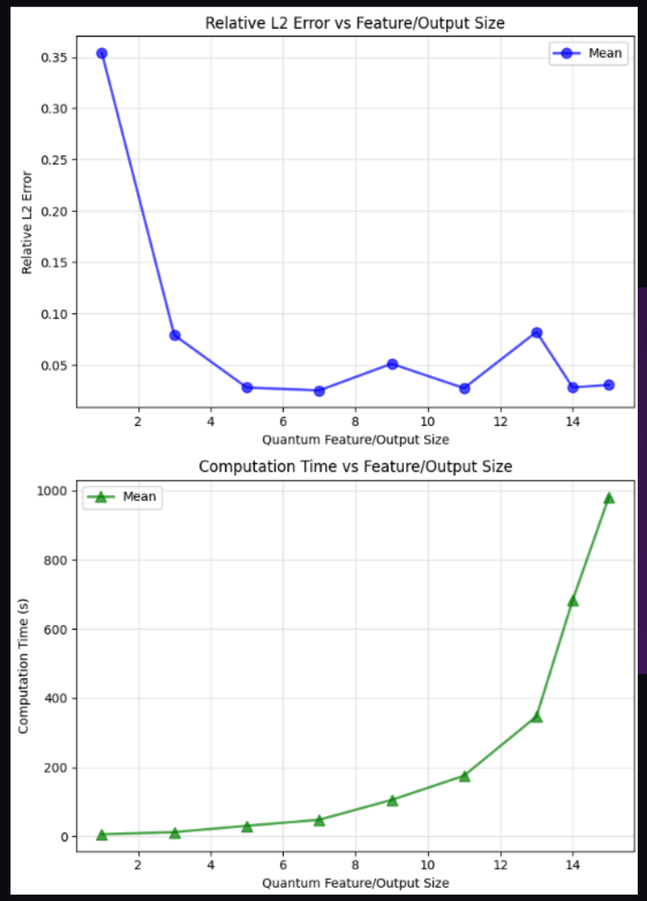
1. Quantum-resource ablation
    a. Surgical ablation
    b. Retrained ablation
2. Classical-simulatibility scaling: can we efficiently simulate it classically ?

# Results: Are there real quantum effects?

Mode mixing, Phase shift, HOM
We can simulate the baseline classicaly, but...

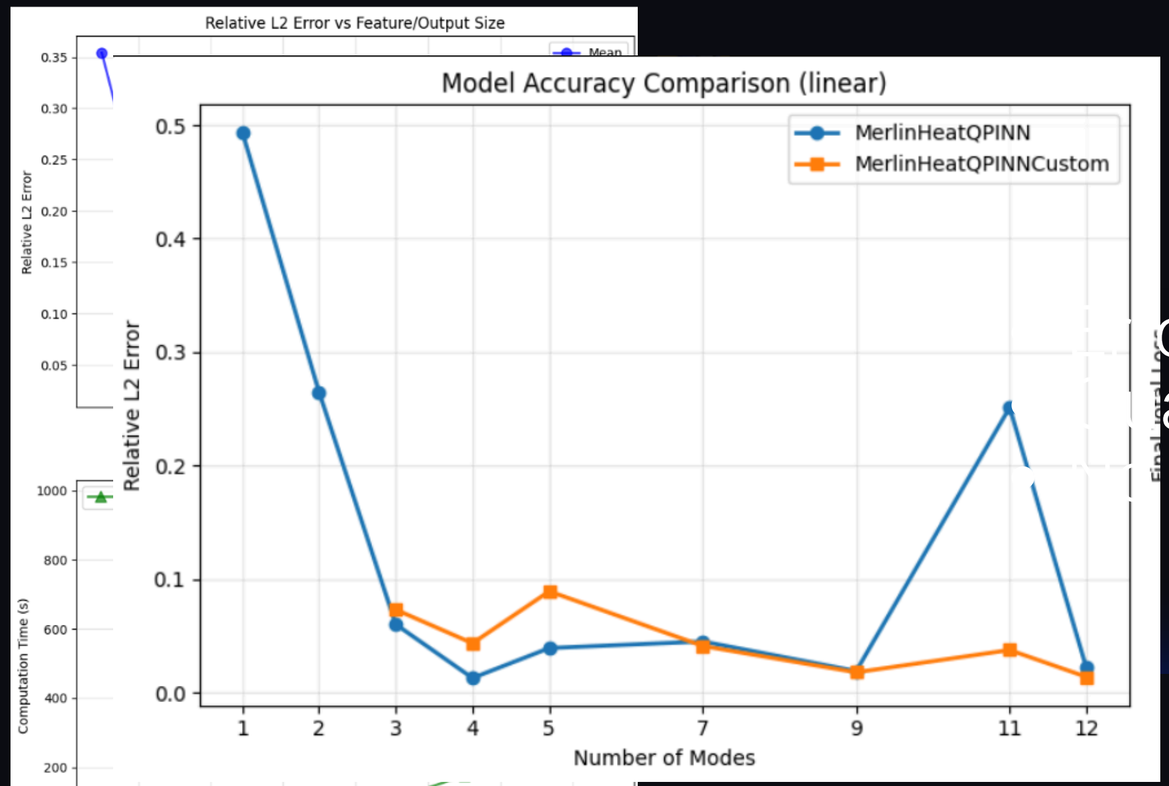
## We see degradation of RMSE

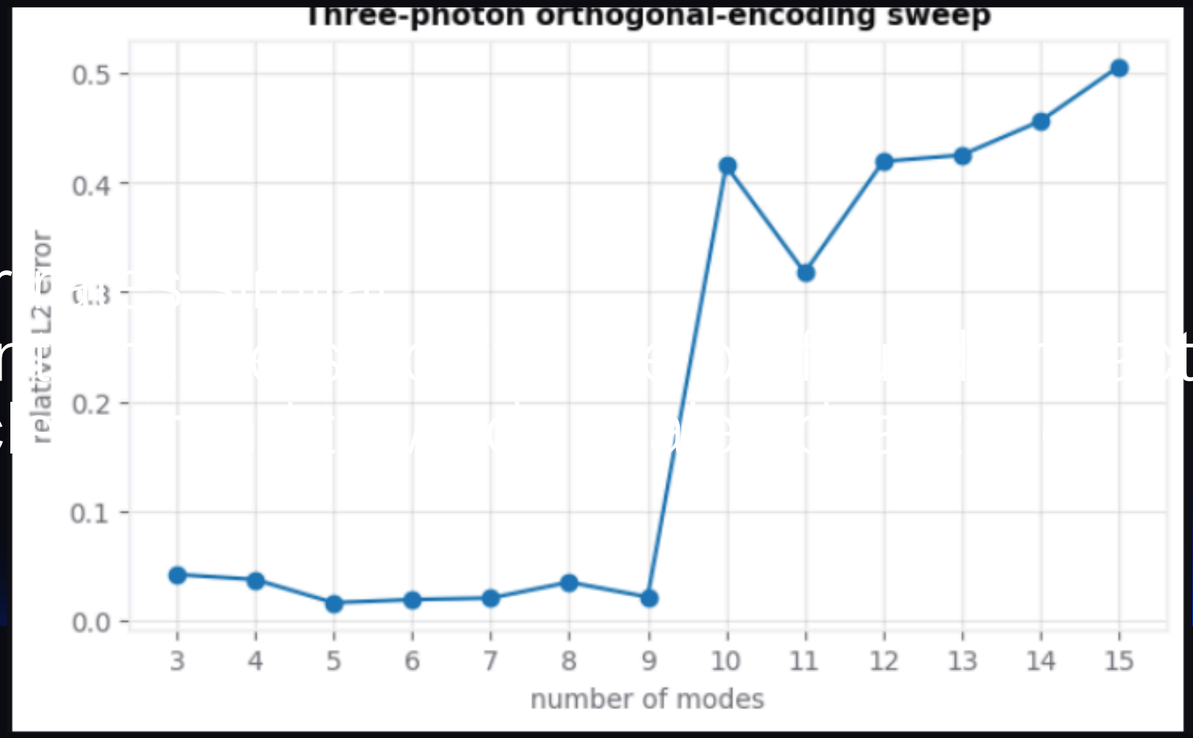
# Results: What about quantum advantage?




- Error rates similar
- Quantum layers don’t have profound impact
  - Frozen layer/
- No clear trend towards scale advantage

# Results: What about quantum advantage?

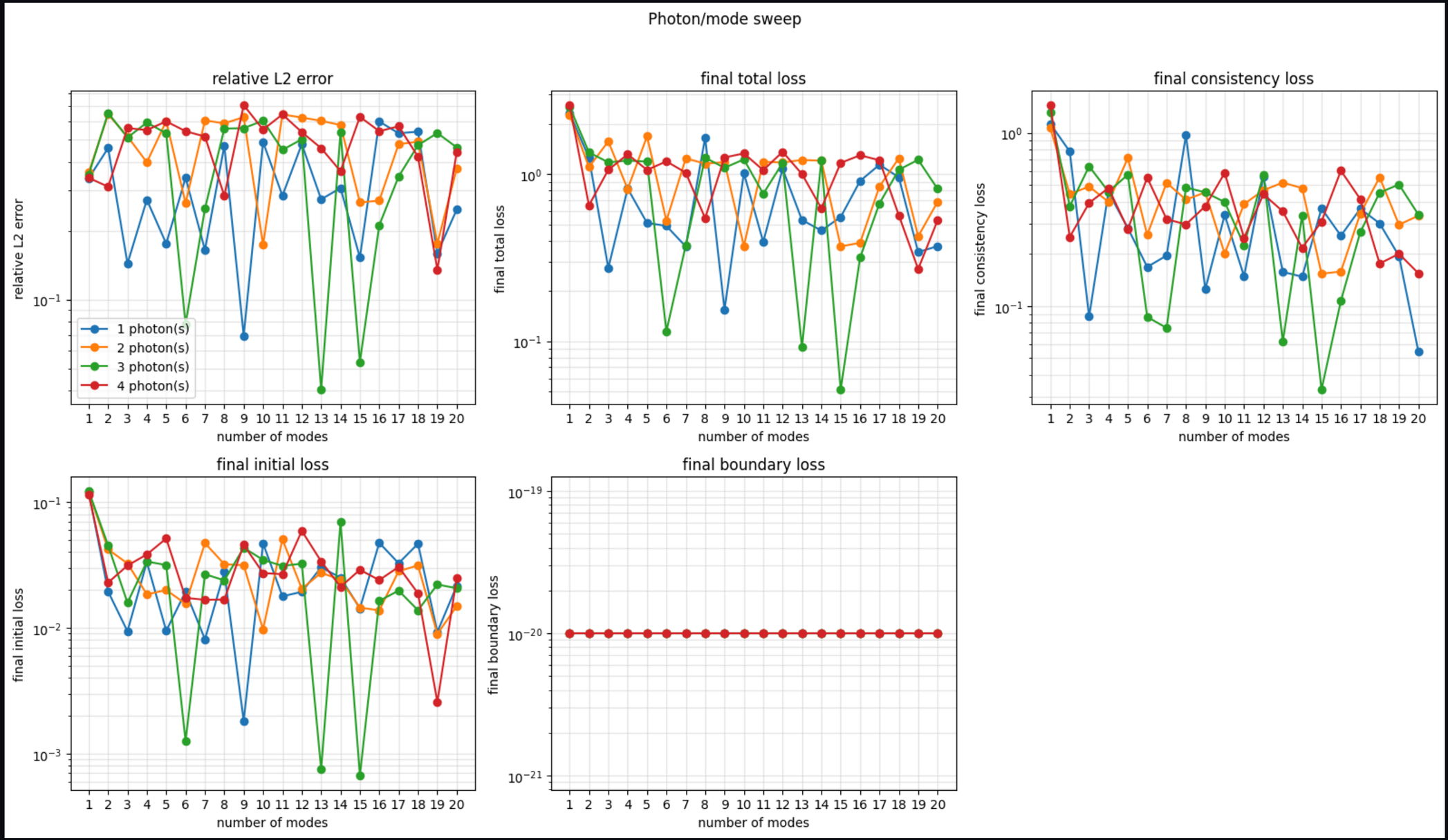


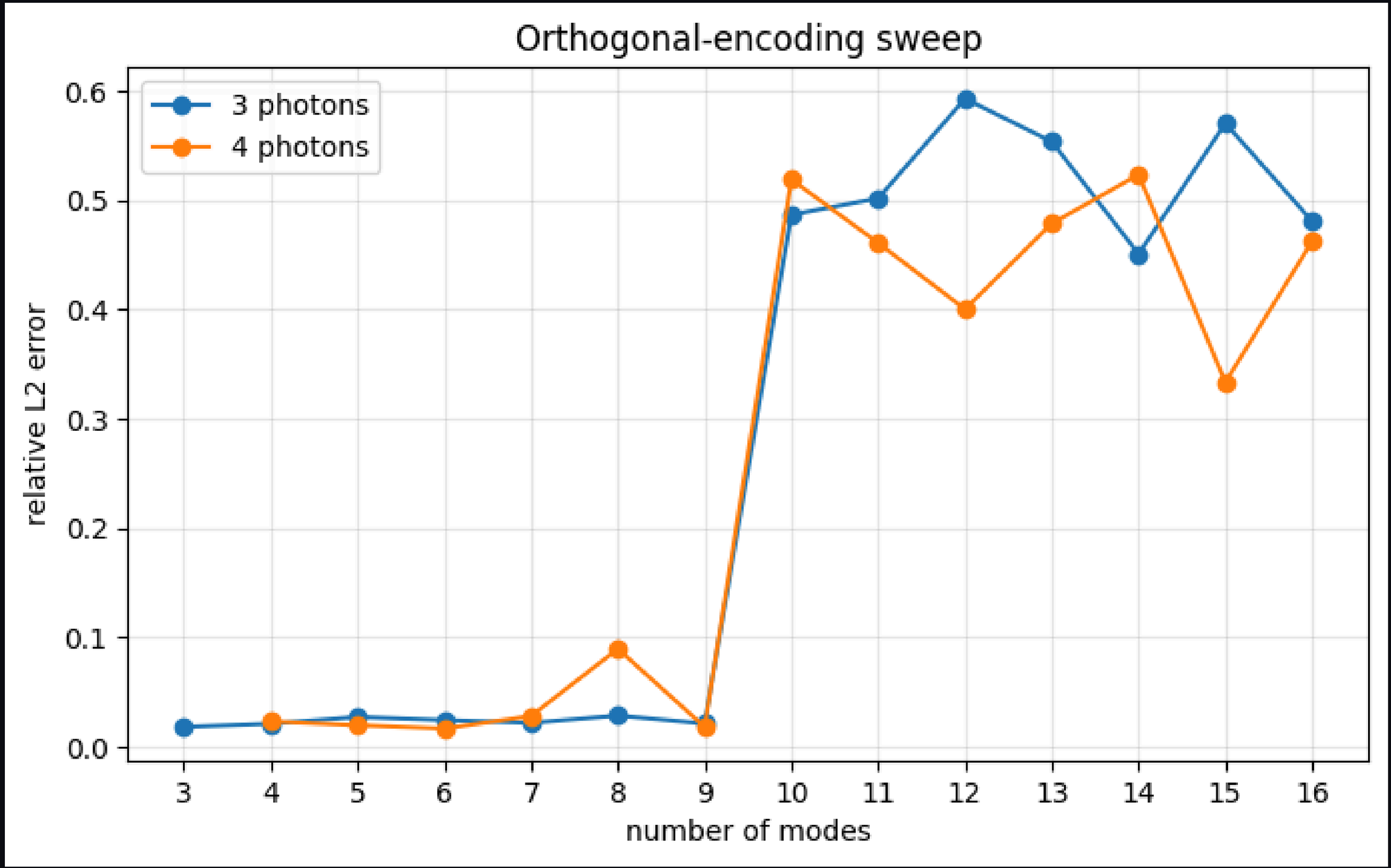



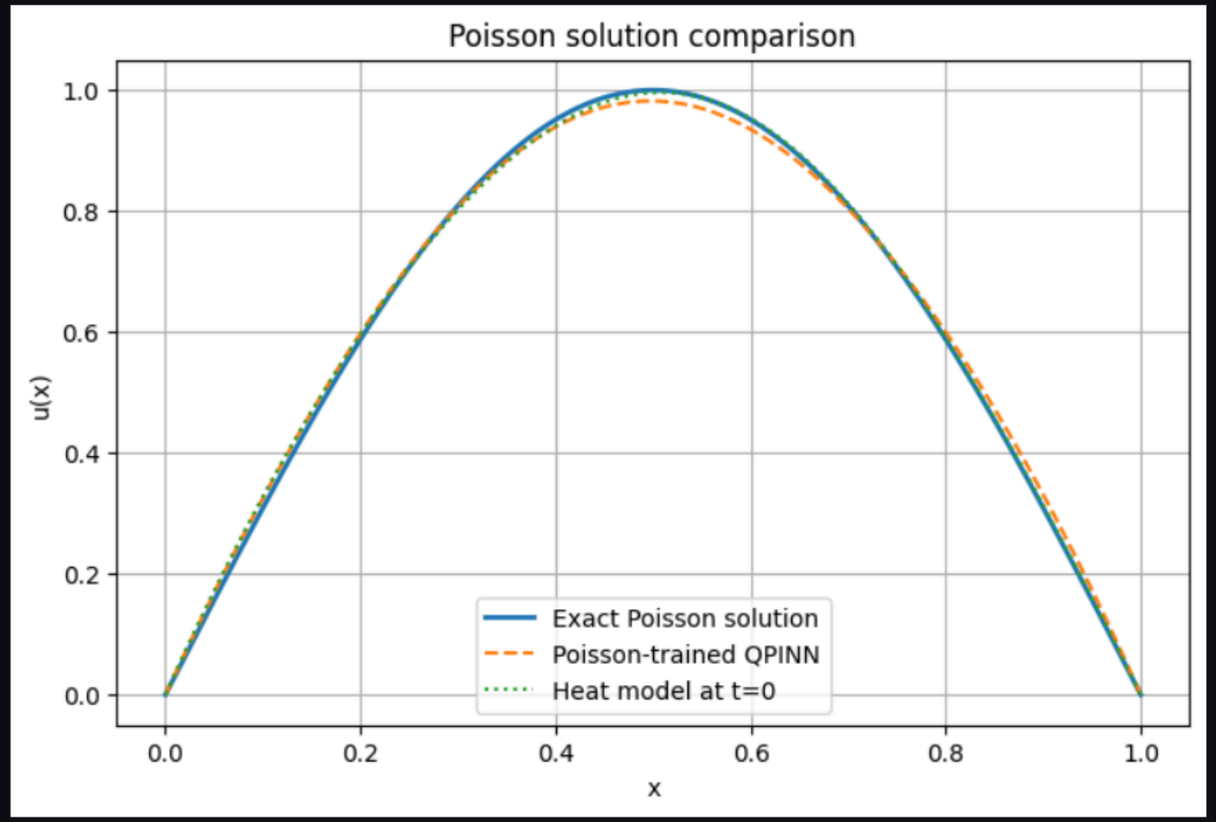
# Thanks !

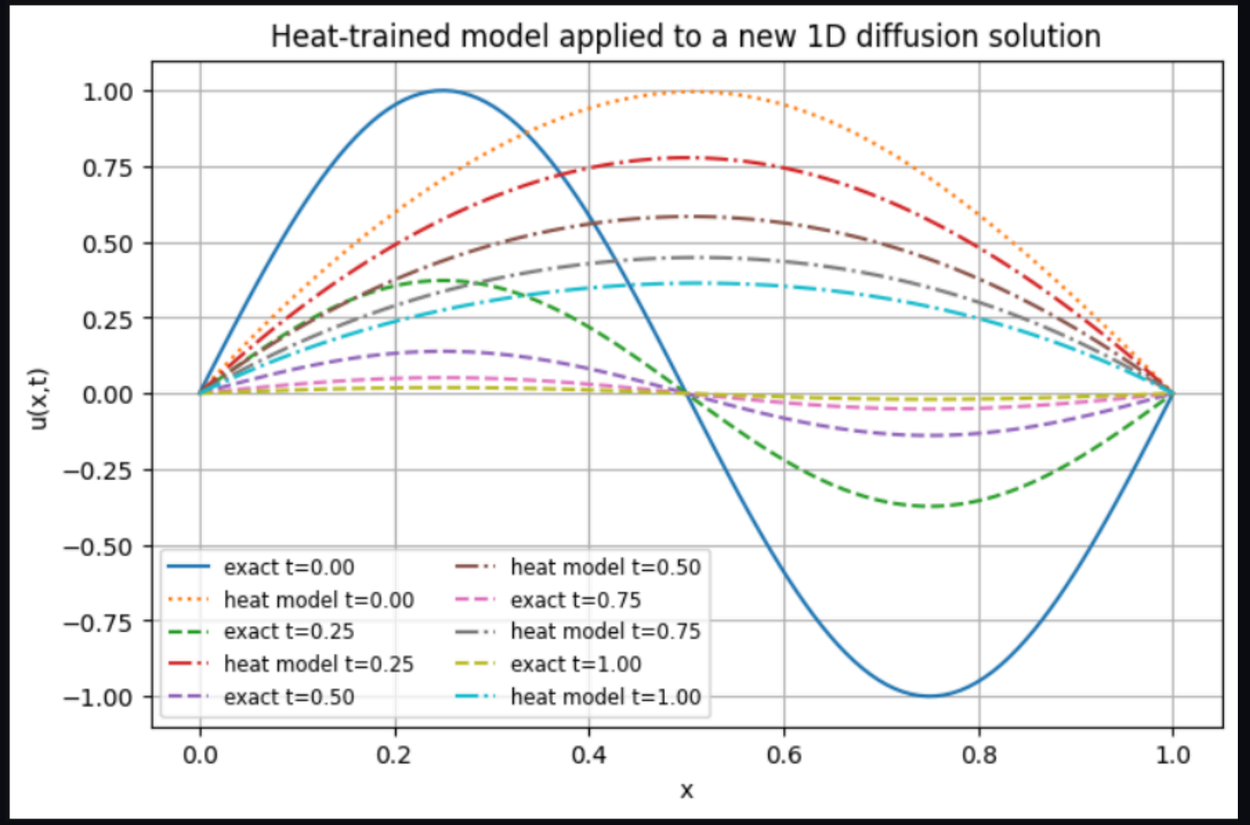
We really enjoyed the challenge and learned a lot :D

# Annexes


Photon/mode sweep
relative L2 error
relative L2 error
number of modes
1 photon(s)
2 photon(s)
3 photon(s)
4 photon(s)
final total loss
final total loss
number of modes
final consistency loss
final consistency loss
number of modes
final initial loss
final initial loss
number of modes
final boundary loss
final boundary loss
number of modes


Orthogonal-encoding sweep
3 photons
4 photons
relative L2 error
0.0
0.1
0.2
0.3
0.4
0.5
0.6
3
4
5
6
7
8
9
10
11
12
13
14
15
16
number of modes


Poisson solution comparison
u(x)
x
Exact Poisson solution
Poisson-trained QPINN
Heat model at t=0


Heat-trained model applied to a new 1D diffusion solution
u(x,t)
x
exact t=0.00
heat model t=0.00
exact t=0.25
heat model t=0.25
exact t=0.50
heat model t=0.50
exact t=0.75
heat model t=0.75
exact t=1.00
heat model t=1.00